

1) Find the coordinates of the foot of the perpendicular drawn from the point $(2, 3, -8)$ to the line $\frac{4-x}{2} = \frac{y}{6} = \frac{1-z}{3}$

2) Find the shortest distance between the lines L_1 and L_2 .

L_1 : The line passing through $(2, -1, +1)$ and parallel to $\frac{x}{1} = \frac{y}{1} = \frac{z}{3}$

$$L_2: \vec{r} = \hat{i} + (2\mu+1)\hat{j} + (-\mu-2)\hat{k}$$

3) Solve the LPP graphically:

$$\text{Maximise } Z = 60x + 40y$$

Subject to

$$x + 2y \leq 12$$

$$2x + y \leq 12$$

$$4x + 5y \geq 20$$

$$x, y \geq 0$$

4) Let S be a relation defined by

$$S = \{(l_1, l_2) : l_1 \text{ is perpendicular to } l_2\}$$

Check whether the relation S is symmetric and Transitive.

5) Sand is pouring from a pipe at the rate of $15 \text{ cm}^3/\text{minute}$. The falling sand forms a cone on the ground such that the height of the cone is always one-third of the radius of the base. How fast is the height of the sand cone increasing at the instant when the height is 4 cm ?

6) The value of $-\int_1^e \log x \, dx$ is